COL703|7203

22nd October

Predicate logic

Syntax

+ variables

Terms →

signature Σ

- constant symbols
- function symbols
- predicate symbols
- propositions (atoms)

nullary

constant
variable

$t_1 \ldots t_n$ terms $f(t_1 \ldots t_n)$

Formulas → $P(t_1 \ldots t_n)$ is a formula

$\phi \wedge \psi$, $\vee$, $\rightarrow$ $\forall x \phi$ $\neg \phi$ $\exists x \phi$

Natural deduction

forall

exists

free & bound variables

Substitution

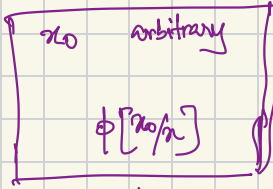
$\phi[t/x]$

$\dfrac{\forall x \phi}{\phi[t/x]}$

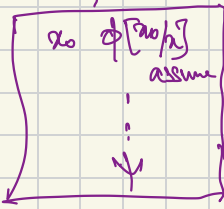
$\dfrac{\phi[t/x]}{\exists x \phi}$

$\dfrac{}{t = t}$

$\dfrac{t_1 = t_2 \quad \phi[t_1/x] \quad \psi}{\phi[t_2/x]}$



$\forall x \phi$



$\exists x \phi$

Semantics

$\mathcal{S}$ - structure (given a signature σ)

assignment / model ($\mathcal{A}$)

non-empty universe $U_{\mathcal{A}}$

for each k-ary predicate P

$P_{\mathcal{A}} \subseteq U_{\mathcal{A}} \times U_{\mathcal{A}} \times \ldots \times U_{\mathcal{A}}$ (k times)

for each k-ary function symbol f

$f_{\mathcal{A}} : U_{\mathcal{A}} \times \ldots \times U_{\mathcal{A}} \to U_{\mathcal{A}}$ (k times)

every constant should be mapped to an element in the universe

every variable is mapped to an element in the universe.

each atomic prop has to be mapped to T/F.

Given an assignment A

In order to evaluate formulas we first need to evaluate terms.

$$A[\![c]\!] = c_A$$

$$A[\![x]\!] = x_A$$

$$A[\![f(t_1 .. t_k)]\!] = f_A(A[\![t_1]\!], A[\![t_2]\!], \ldots, A[\![t_k]\!])$$

$A \models P(t_1 \dots t_n)$ if

$(A[\![t_1]\!], A[\![t_2]\!], \dots A[\![t_n]\!]) \in P_A$

$A \models \phi \vee \psi$ $A \models \phi$ or $A \models \psi$

$\phi \wedge \psi$

$\phi \to \psi$

$A \models \neg\phi$ $A \not\models \phi$

$A \models \exists x \phi$ if $\exists a \in U_A$ $A_{[x \mapsto a]} \models \phi$

$A \models \forall x \phi$ if $\forall a \in U_A$ $A_{[x \mapsto a]} \models \phi$

Same as A except that x is mapped to a.

$A \models t_1 = t_2$ iff $A[\![t_1]\!] = A[\![t_2]\!]$

Assignment really has two parts



structure / interpretation

universe

$\mathcal{S}$

variable assignment

ii) equality is not open for interpretation

iii) atomic prop's get a T/F value

part of structure / interpretation (not variable assignment)

$$P(0) \wedge \forall x \left(P(x) \rightarrow P(S(x)) \right) \wedge \exists x \neg P(x)$$

signature { constant symbol 0, unary function S, unary predicate P }

Universe: all non-negative integers

$S(x) = x$

$P = \{0\}$

0 (counting) is mapped to 0 (in the numbers)

A formula with no free variables is called a sentence (it is also called closed).

$$\exists x \exists z \left(P(x) \wedge \forall y \; R(x, y, z) \right)$$

existential closure

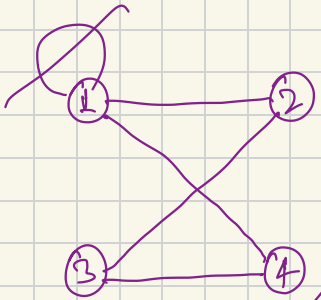
open formula

AFΦ

undirected graph

& structure

$E(x,y)$



Universe $= \{1, 2, 3, 4\}$

$E = \{(1,2), (2,3), (3,4), (4,1), (2,1), (3,2), (4,3), (1,4)\}$ ~~$(1,1)$~~

$\forall x \neg E(x,x)$

$\wedge \forall x \forall y (E(x,y) \rightarrow E(y,x))$

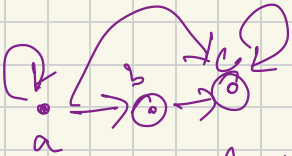
$$\psi \quad \forall x \forall y \exists z_1 \exists z_2 \Big(E(x, z_1) \wedge E(z_1, z_2) \wedge E(z_2, y) \Big)$$

$$A \not\models \psi$$

Example

Consider a signature σ

i	constant symbol
F	unary predicate
P	binary predicate



Consider the following structure / interpretation A

universe $= \{a, b, c\}$

constant i is mapped to a

$F = \{b, c\}$

$P = \{(a,a), (a,b), (a,c), (b,c), (c,c)\}$

$\exists y\ P(i,y)$

ϕ

$A \models \phi$

$\neg F(i)$

$\forall x \forall y \forall z \left(P(x,y) \wedge P(x,z) \rightarrow y = z \right)$

not true in the given structure

$\forall x \exists y \; P(x,y)$

Example

Signature σ

a constant symbol e

a binary fun symbol $\cdot$

a binary pred. $\leq$

σ-structure

universe : all finite words over $\{0,1\}$ along with ϵ

e maps to ϵ

$\cdot$ is concatenation $\cdot(x,y) = x \cdot y$

$\leq$ is the prefix ordering $\leq(s_1, s_2)$ is true is s_1 is a prefix of s_2.

$$\forall x \left((x \leq x \cdot e) \wedge (x \cdot e \leq x) \right)$$

$$\exists y \, \forall x \, (y \leq x)$$

$$\forall x \, \exists y \, (y \leq x)$$

$$\forall x \, \forall y \, \forall z \left((x \leq y) \rightarrow (x \cdot z \leq y \cdot z) \right)$$

$01 \quad 011 \qquad 110 \quad 10 \qquad 010 \not\leq 0110$

Exercise

$$\neg \exists x \forall y \left((x \leq y) \rightarrow (y \leq x) \right)$$

Satisfiability, Validity > same as in prop. logic